

Implicit Objects in JSP

- These objects are created by the web container that are available to all the jsp pages.
- These objects are also called as Pre-Defined Variables in JSP.
- JSP implicit objects are created during the translation phase of JSP to the servlet.
- There are **9 types** of implicit objects available in the JSP :

1.out

2.request

3.response

4.session

5.exception

6.application

7.pageContext

8.page

9.config

Out Object:

- **Out** is one of the implicit objects to write the data to the buffer and send output to the client(browser) in response.
- It is the object of **JspWriter** Class.

Example: “**outobj.jsp**”

```
<html>
```

```
<body>
```

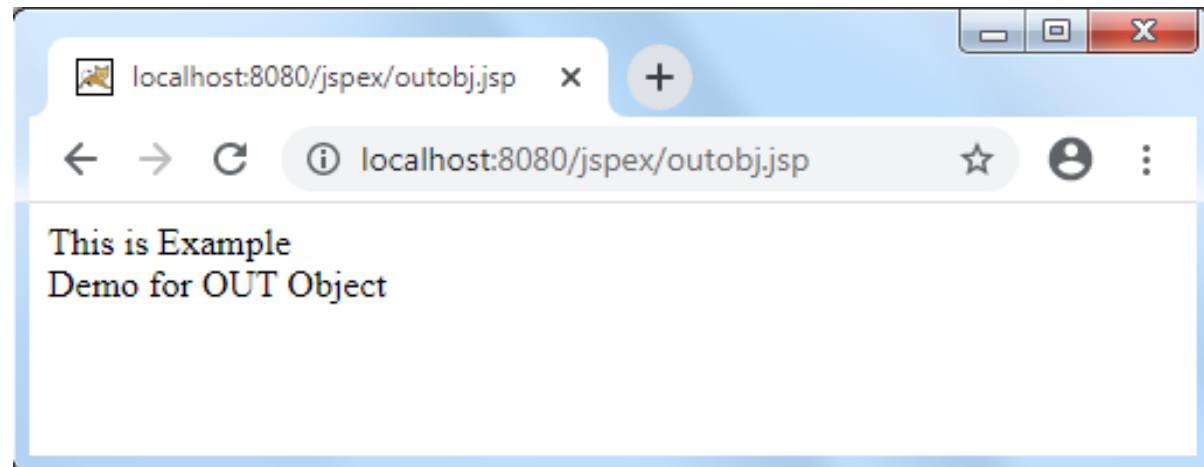
```
<% out.print("This is Example <br/> ");
```

```
out.print(" Demo for OUT Object");
```

```
%>
```

```
</body>
```

```
</html>
```



Request Object:

- The **request** is an implicit object of type `HttpServletRequest`. It was created for each jsp request by the web container.
- It can be used to get request information from web page or document.
- By using this **request** object ,we can able read values from input elements in JSP.

Example:

“requestobj.html”

```
<html>
<head>
<title>Request Object</title>
</head>
<body>
<form action="requestobj.jsp">
<input type="text" name="uname">
<input type="submit" value="go"><br/>
</form>
</body>
</html>
```

requestobj.jsp

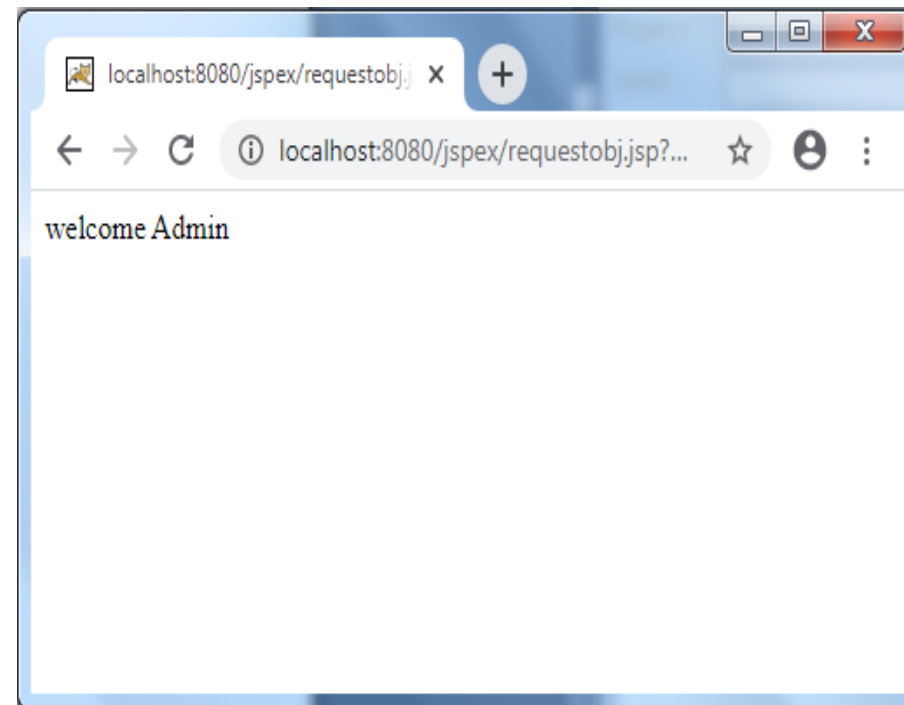
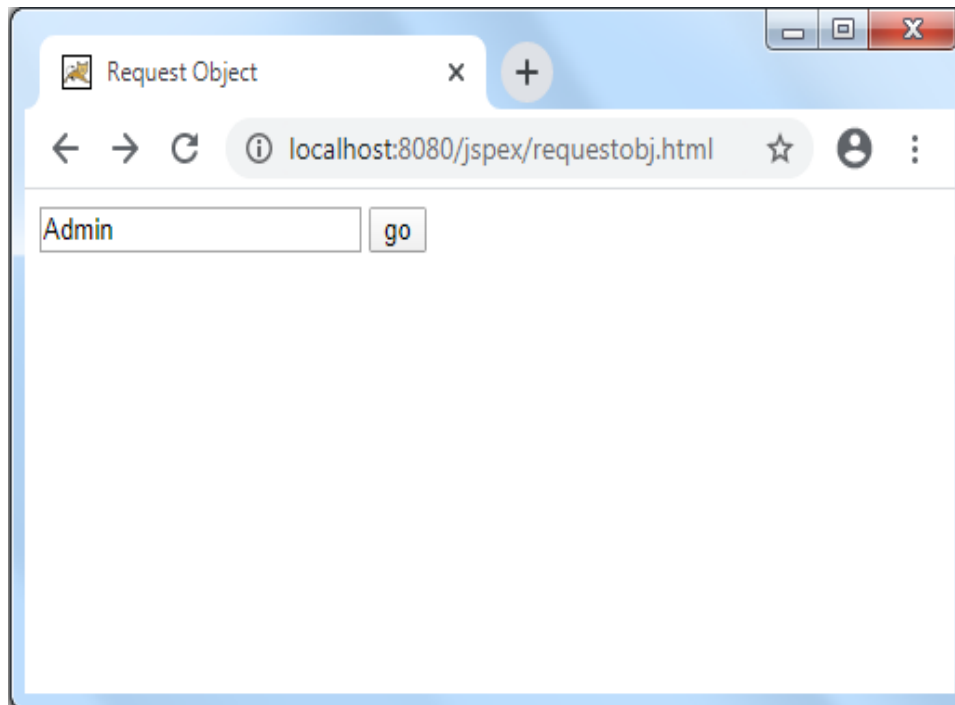
<%

String name=request.getParameter("uname");

out.print("welcome "+name);

%>

Output:



Response Object:

- In JSP, **response** is an implicit object of type HttpServletResponse. It is created by the web container for each jsp request.
- It can be used to add or manipulate response such as redirect response to another resource, send error etc.

Example:

```
<html>
<head>
<title>Response Object</title>
</head>
<body>
<form action="responseobj.jsp">
URL:<input type="text" name="url">
<input type="submit" value="go"><br/>
</form>
</body>
</html>
```

responseobj.jsp

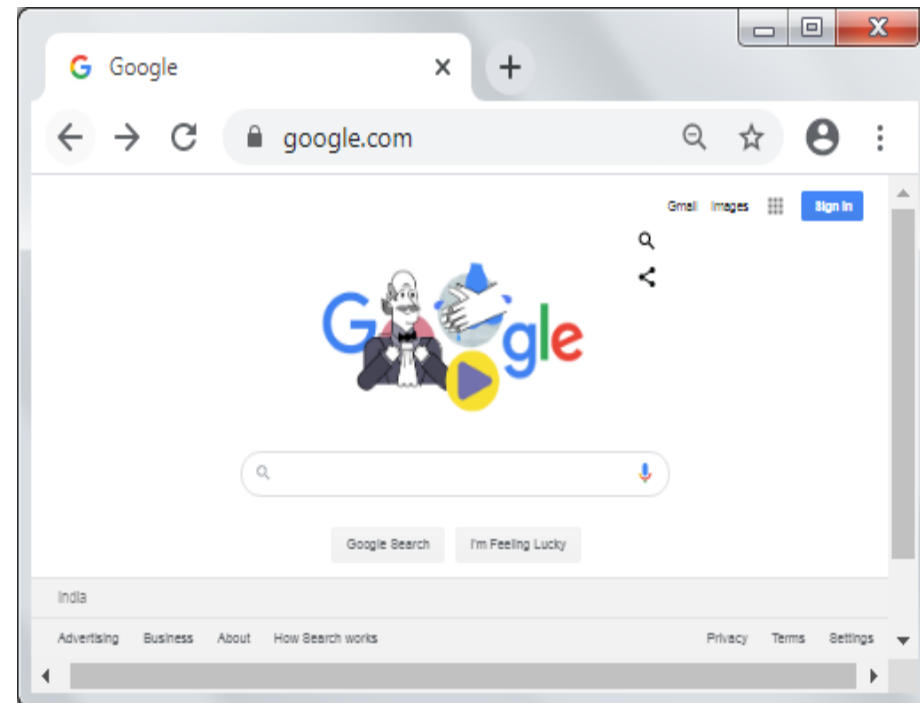
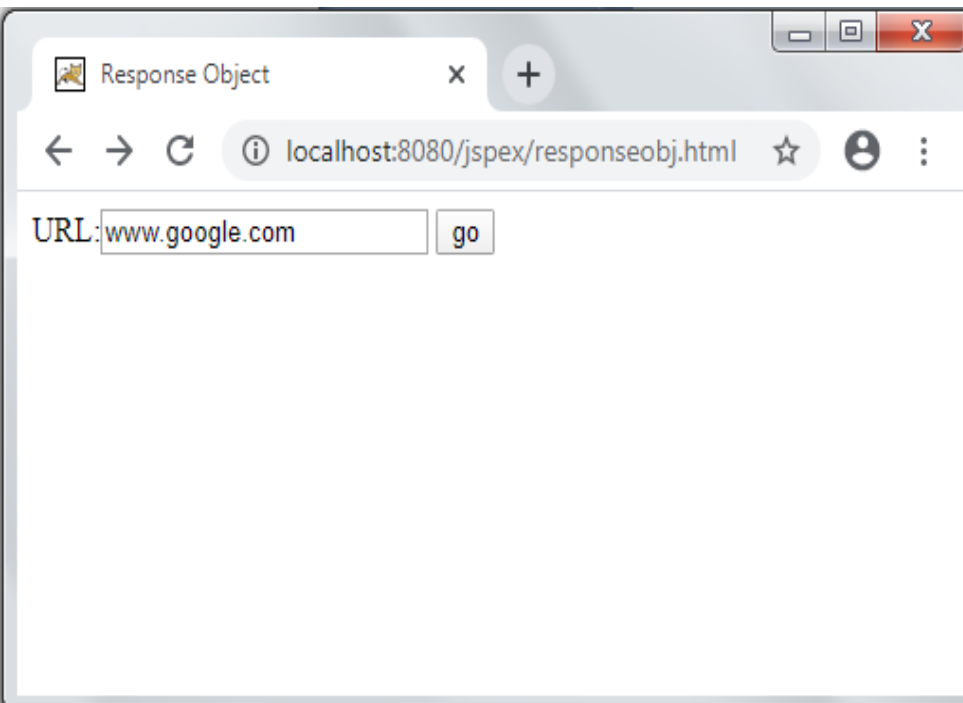
`<%`

`String url=request.getParameter("url");`

`response.sendRedirect("http://" + url);`

`%>`

Output:



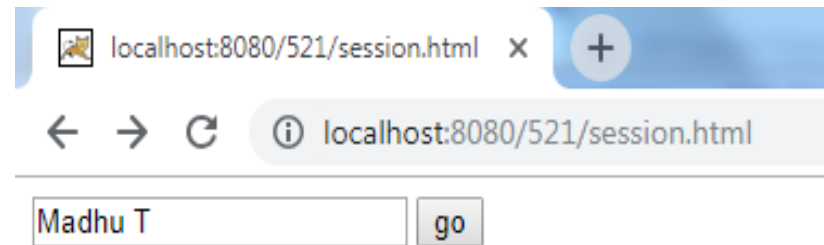
Session Object:

- In JSP, session is an implicit object of type HttpSession. We can use this object to set, get or remove attribute or to get session information.

Example:

Session.html

```
<html>
<body>
<form action="welcome.jsp">
<input type="text" name="uname">
<input type="submit" value="go">
</form>
</body>
</html>
```



session.jsp

```
<html>
```

```
<body>
```

```
<%
```

```
String name=request.getParameter("uname");
```

```
out.print("Welcome "+name);
```

```
session.setAttribute("user",name);
```

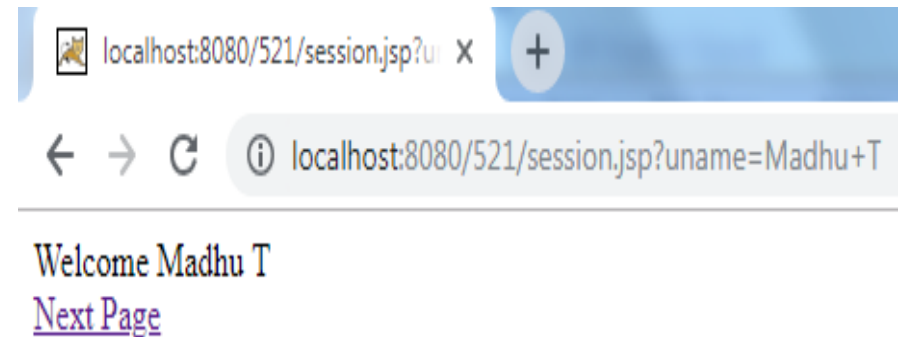
```
%>
```

```
<br/>
```

```
<a href="second.jsp">Next Page</a>
```

```
</body>
```

```
</html>
```



second.jsp

```
<html>
```

```
<body>
```

```
<%
```

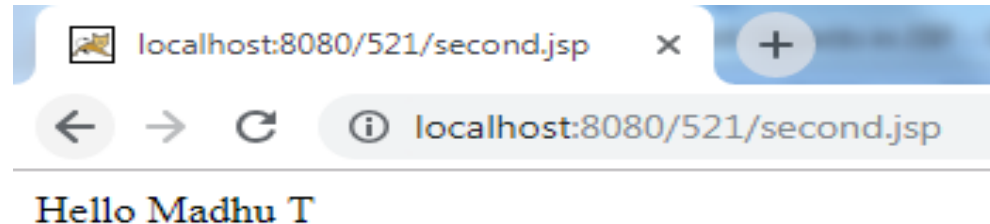
```
String name=(String)session.getAttribute("user");
```

```
out.print("Hello "+name);
```

```
%>
```

```
</body>
```

```
</html>
```



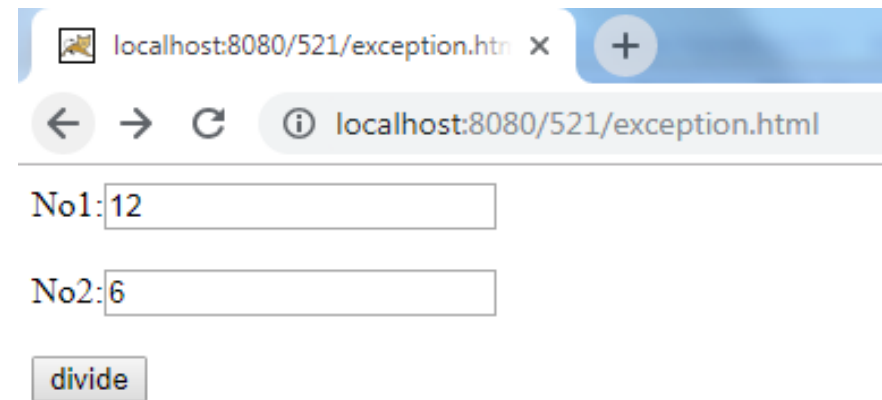
Exception Object:

- The exception is normally an object that is thrown at runtime. Exception Handling is the process to handle the runtime errors. There may occur exception any time in your web application.
- In JSP, exception is an implicit object of type `java.lang.Throwable` class. This object can be used to print the exception. But it can only be used in error pages.

Example:

exception.html

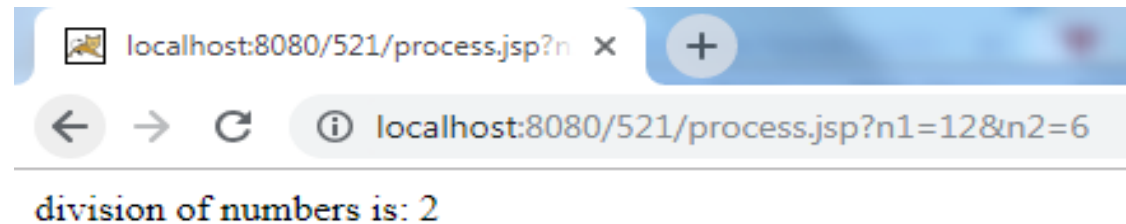
```
<html>
<body>
<form action="process.jsp">
No1:<input type="text" name="n1" />
No2:<input type="text" name="n2" />
<input type="submit" value="divide"/>
</form>
</body>
</html>
```



The screenshot shows a web browser window with the address bar displaying 'localhost:8080/521/exception.html'. The page content includes two text input fields: 'No1' containing the value '12' and 'No2' containing the value '6'. Below these fields is a button labeled 'divide'.

process.jsp

```
<%@ page errorPage="error.jsp" %>
<%
String num1=request.getParameter("n1");
String num2=request.getParameter("n2");
int a=Integer.parseInt(num1);
int b=Integer.parseInt(num2);
int c=a/b;
out.print("division of numbers is: "+c);
%>
```



error.jsp

```
<html>
```

```
<body>
```

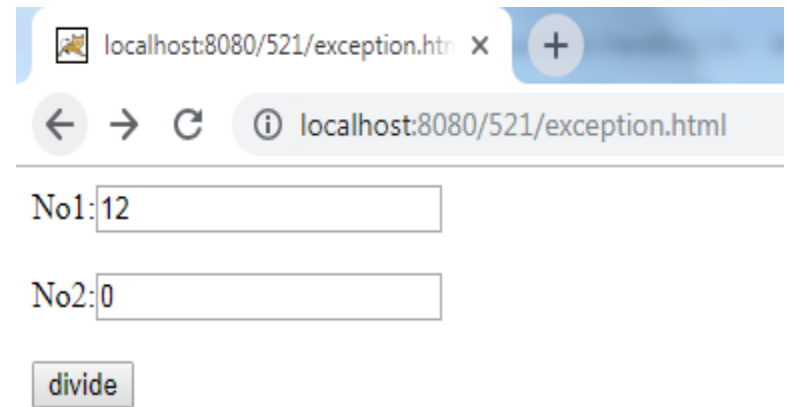
```
<%@ page isErrorPage="true" %>
```

```
<h3>Sorry an exception occurred!</h3>
```

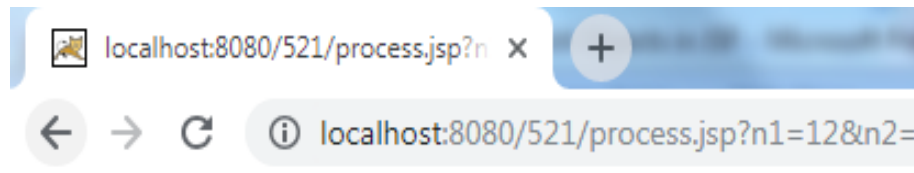
```
Exception is: <%= exception %>
```

```
</body>
```

```
</html>
```



A screenshot of a web browser window. The address bar shows 'localhost:8080/521/exception.html'. The page contains two input fields: 'No1:' with the value '12' and 'No2:' with the value '0'. Below these fields is a button labeled 'divide'.



A screenshot of a web browser window. The address bar shows 'localhost:8080/521/process.jsp?n1=12&n2=0'. The page displays the message 'Sorry an exception occurred!' and 'Exception is: java.lang.ArithmeticException: / by zero'.

Sorry an exception occurred!

Exception is: java.lang.ArithmeticException: / by zero

Application Object:

- This is basically used for getting initialization parameters and for sharing the attributes & their values across the entire JSP application.
- which means any attribute set by **application implicit object** would be available to all the JSP pages.
- **getAttribute(String attributeName):** It returns the object stored in a given attribute name.

Ex:String s = (String)application.getAttribute("MyAttr");

- **setAttribute(String attributeName, Value):** It sets the value of an attribute or in other words it stores an attribute and its value in application context, which is available to use across JSP application.

Ex:application.setAttribute("MyAttr", "value");

applnobj.jsp

```
<html>
```

```
<%
```

```
Integer counter= (Integer)application.getAttribute("numOfVisits");
```

```
if( counter ==null || counter == 0 ){
```

```
    counter = 1;
```

```
}else{
```

```
    counter = counter+ 1;
```

```
}
```

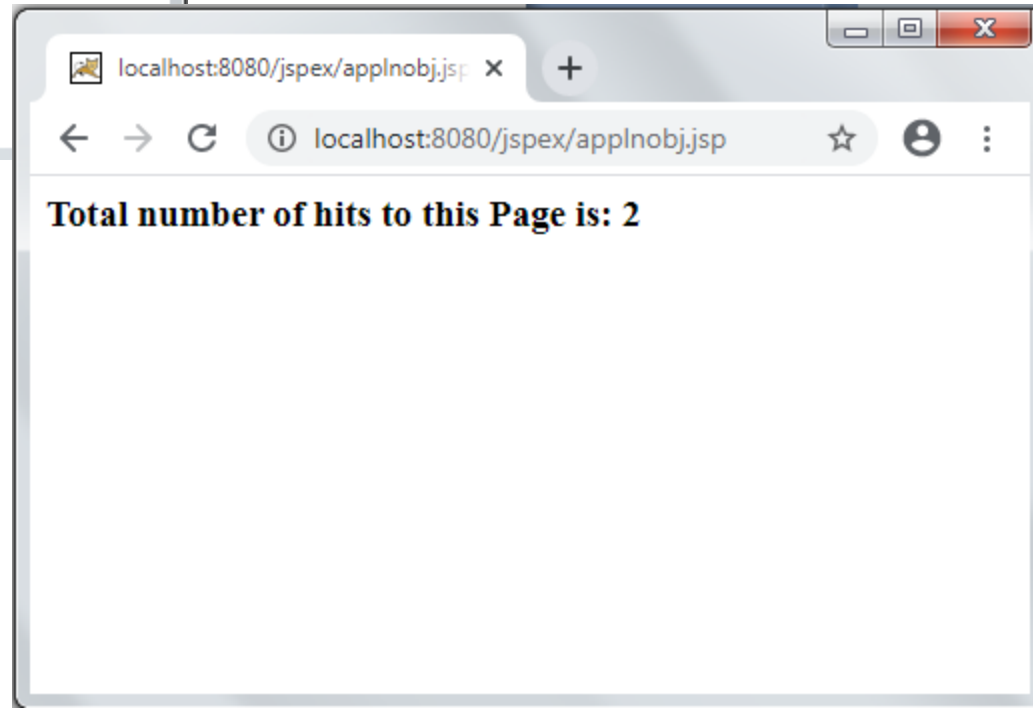
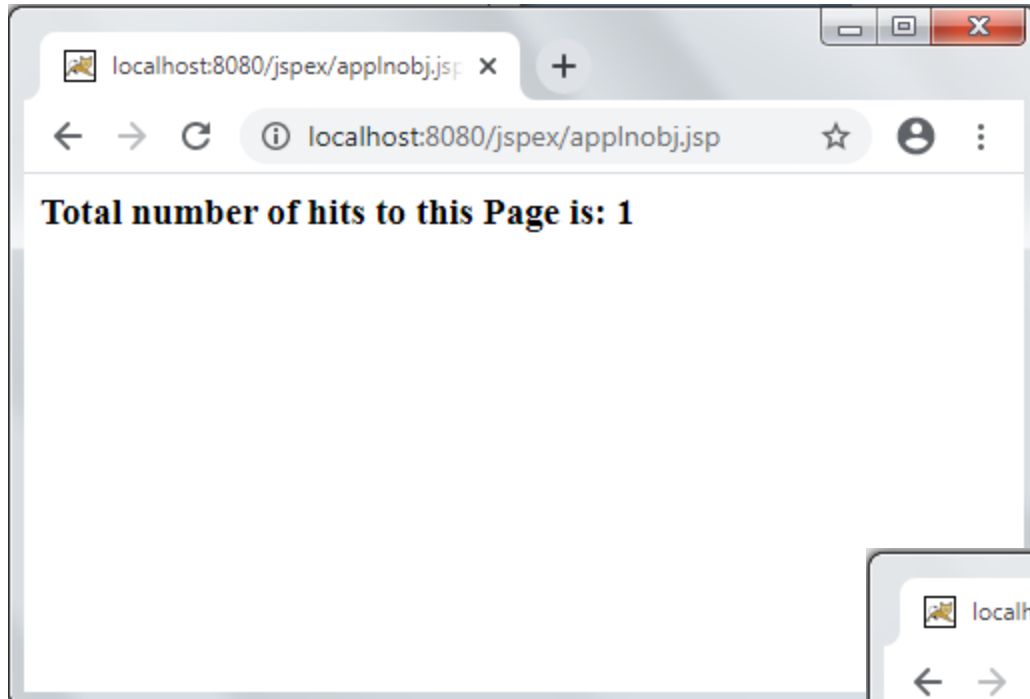
```
application.setAttribute("numOfVisits", counter);
```

```
%>
```

```
<h3>Total number of hits to this Page is: <%= counter%></h3>
```

```
</html>
```

Output:



pageContext Object:

- In JSP, **pageContext** is an implicit object of type PageContext class.
- The **pageContext** object can be used to set, get or remove attribute from one of the following scopes:
 - ✓ **page** – Scope: PAGE_CONTEXT
 - ✓ **request** – Scope: REQUEST_CONTEXT
 - ✓ **session** – Scope: SESSION_CONTEXT
 - ✓ **application** – Scope: APPLICATION_CONTEXT
- In JSP, page scope is the default scope.

Methods of pageContext Implicit Object

- `getAttribute (String AttributeName, int Scope)`
- `setAttribute(String AttributeName, AttributeValue, int Scope)`

pagecontextobj.html

<html>

<head><title> Login Page</title></head>

<body>

<form action="**pagecontextobj.jsp**">

Enter User-Id: <input type="text" name="**uid**">

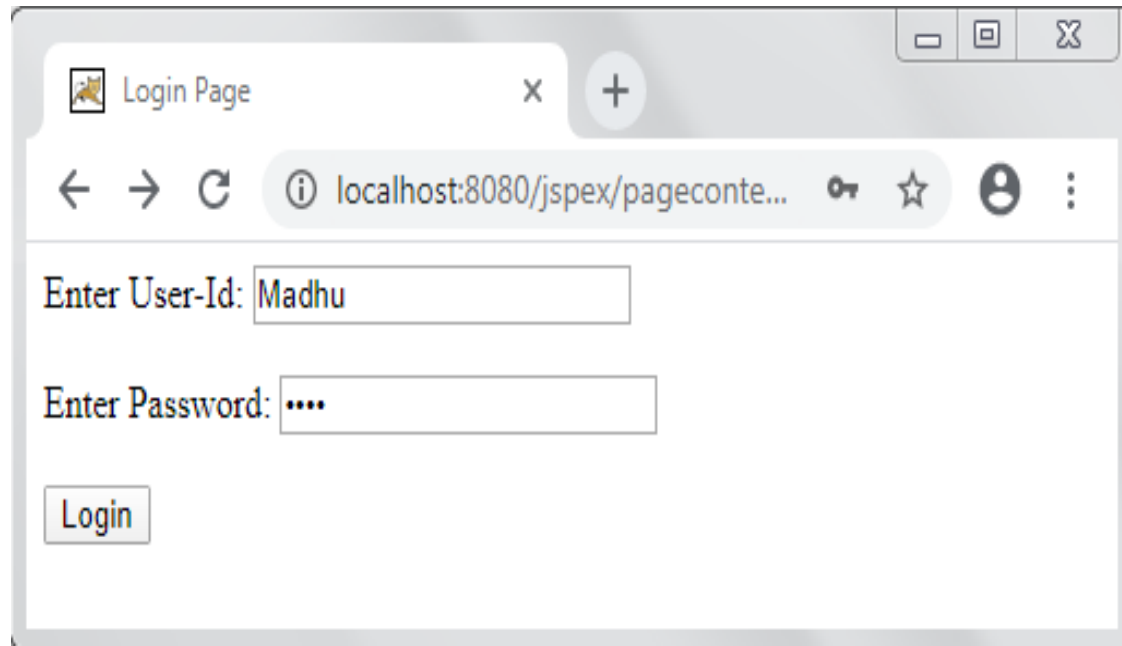
Enter Password: <input type="password" name="**upass**">

<input type="submit" value="Login">

</form>

</body>

</html>



pagecontextobj.jsp

```
<html>
```

```
<body>
```

```
<% String id=request.getParameter("uid");
```

```
String pass=request.getParameter("upass");
```

```
out.println("hello "+id);
```

```
pageContext.setAttribute("uname", id, PageContext.SESSION_SCOPE);
```

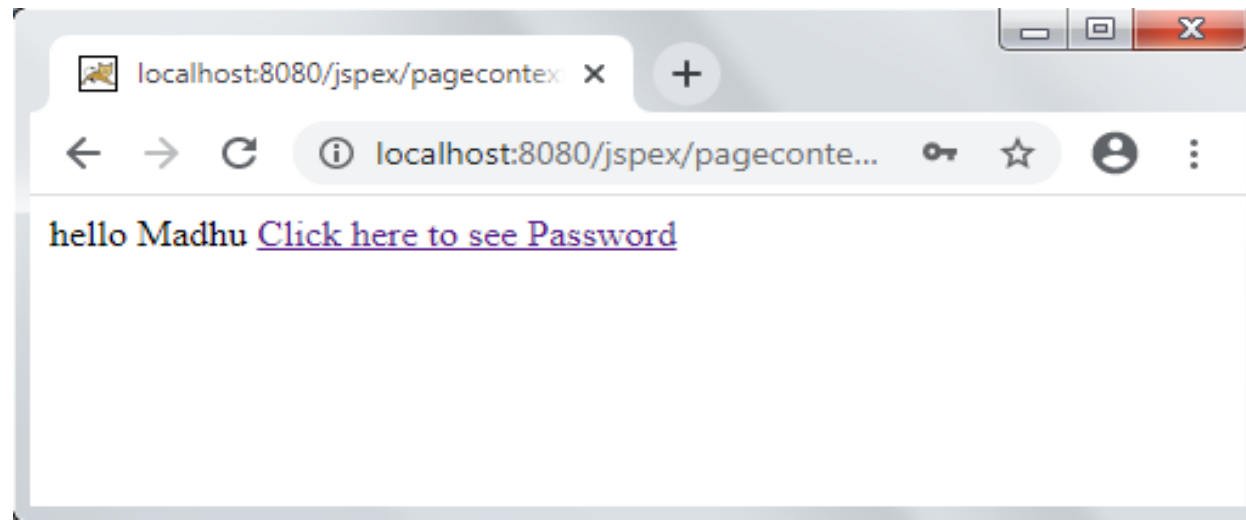
```
pageContext.setAttribute("pwd", pass, PageContext.SESSION_SCOPE);
```

```
%>
```

```
<a href="display.jsp">Click here to see Password </a>
```

```
</body>
```

```
</html>
```



display.jsp

```
<html>
```

```
<body>
```

```
<%
```

```
String uname= (String) pageContext.getAttribute("uname",  
                                                PageContext.SESSION_SCOPE);
```

```
String pwd= (String) pageContext.getAttribute("pwd",  
                                              PageContext.SESSION_SCOPE);
```

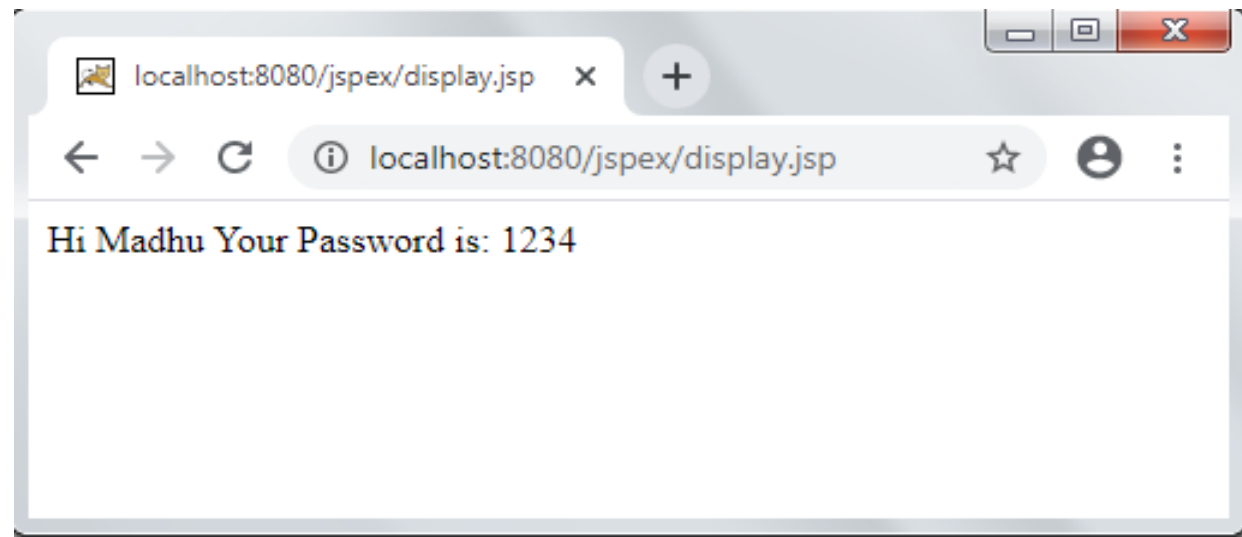
```
out.println("Hi "+uname);
```

```
out.println("Your Password is: "+pwd);
```

```
%>
```

```
</body>
```

```
</html>
```



page Object:

- Page implicit variable holds the currently executed servlet object for the corresponding jsp.
- Acts as this object for current jsp page.
- This object is very rarely used.

Example:

```
<html>
```

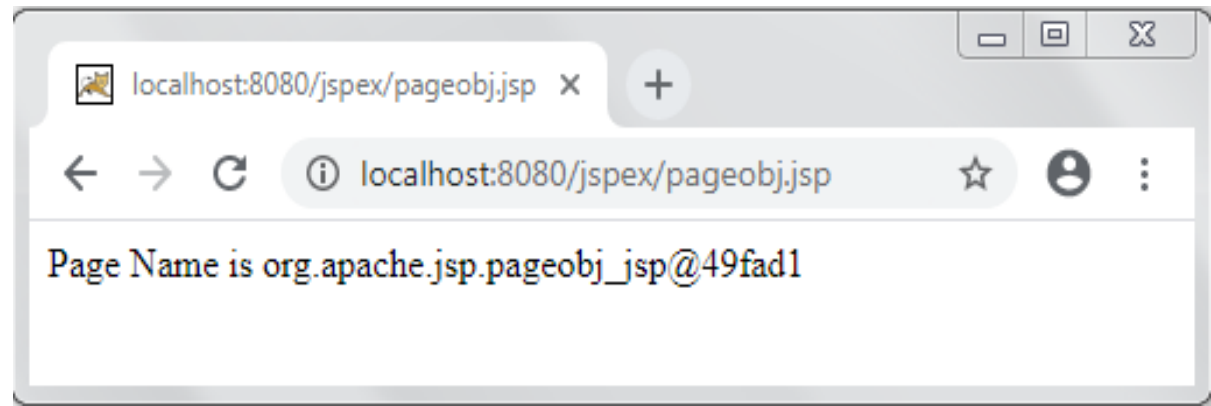
```
<body>
```

```
<% String pageName = page.toString();
```

```
out.println("Page Name is " +pageName);%>
```

```
</body>
```

```
</html>
```



config Object:

- In JSP, config is an implicit object of type ***ServletConfig***. This object can be used to get initialization parameter for a particular JSP page.
- The config object is created by the web container for each jsp page.
- Generally, it is used to get initialization parameter from the web.xml file.

Example:

“configobj.jsp”

```
<html>
```

```
<body>
```

```
<%
```

```
String sname=config.getServletName();
```

```
out.print("Servlet Name is: "+sname);
```

```
%>
```

```
</body>
```

```
</html>
```

web.xml

```
<web-app>
```

```
<servlet>
```

```
<servlet-name>SampleServlet</servlet-name>
```

```
<jsp-file>/configobj.jsp</jsp-file>
```

```
</servlet>
```

```
<servlet-mapping>
```

```
<servlet-name>SampleServlet</servlet-name>
```

```
<url-pattern>/config</url-pattern>
```

```
</servlet-mapping>
```

```
</web-app>
```

